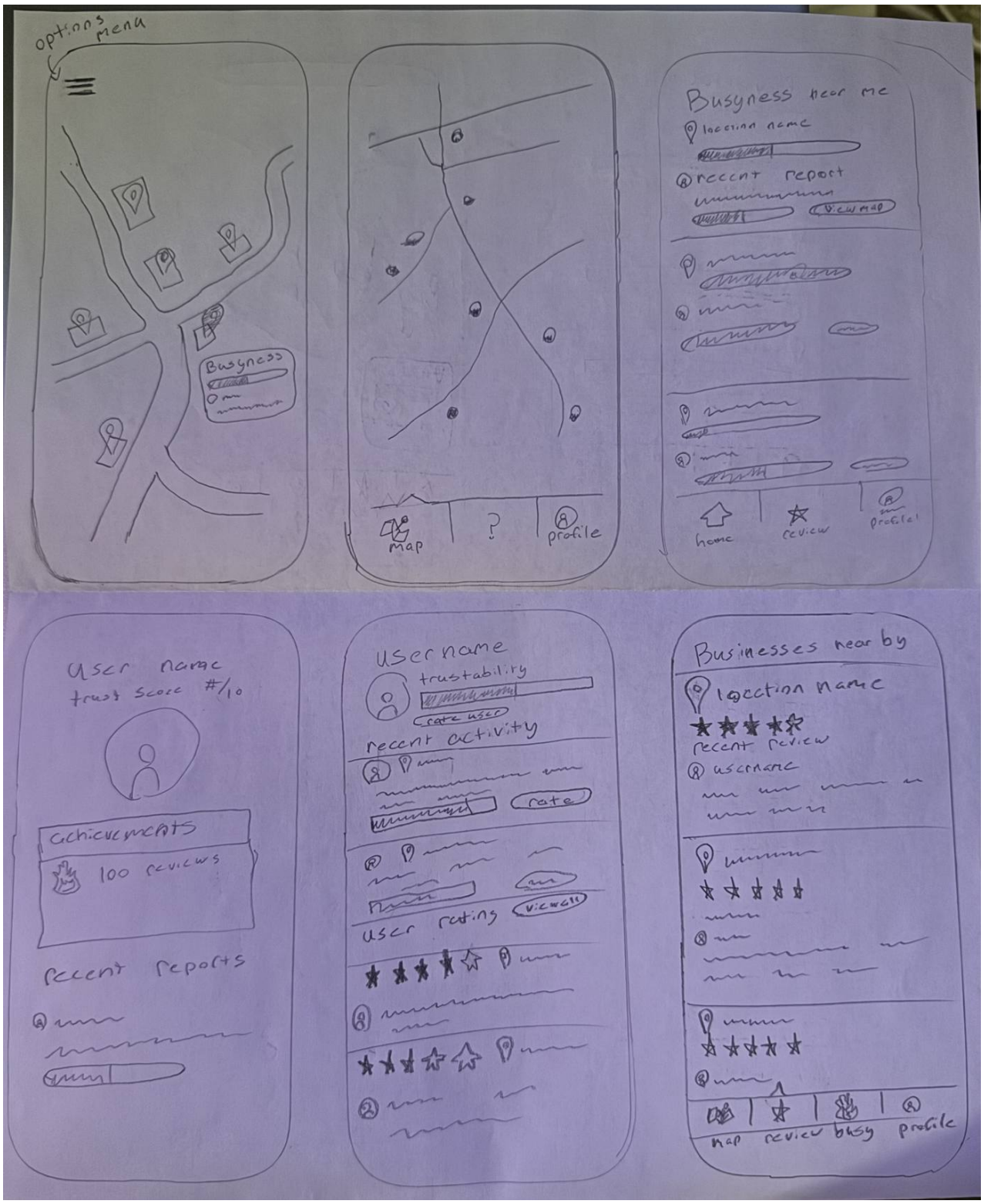
BusyBee

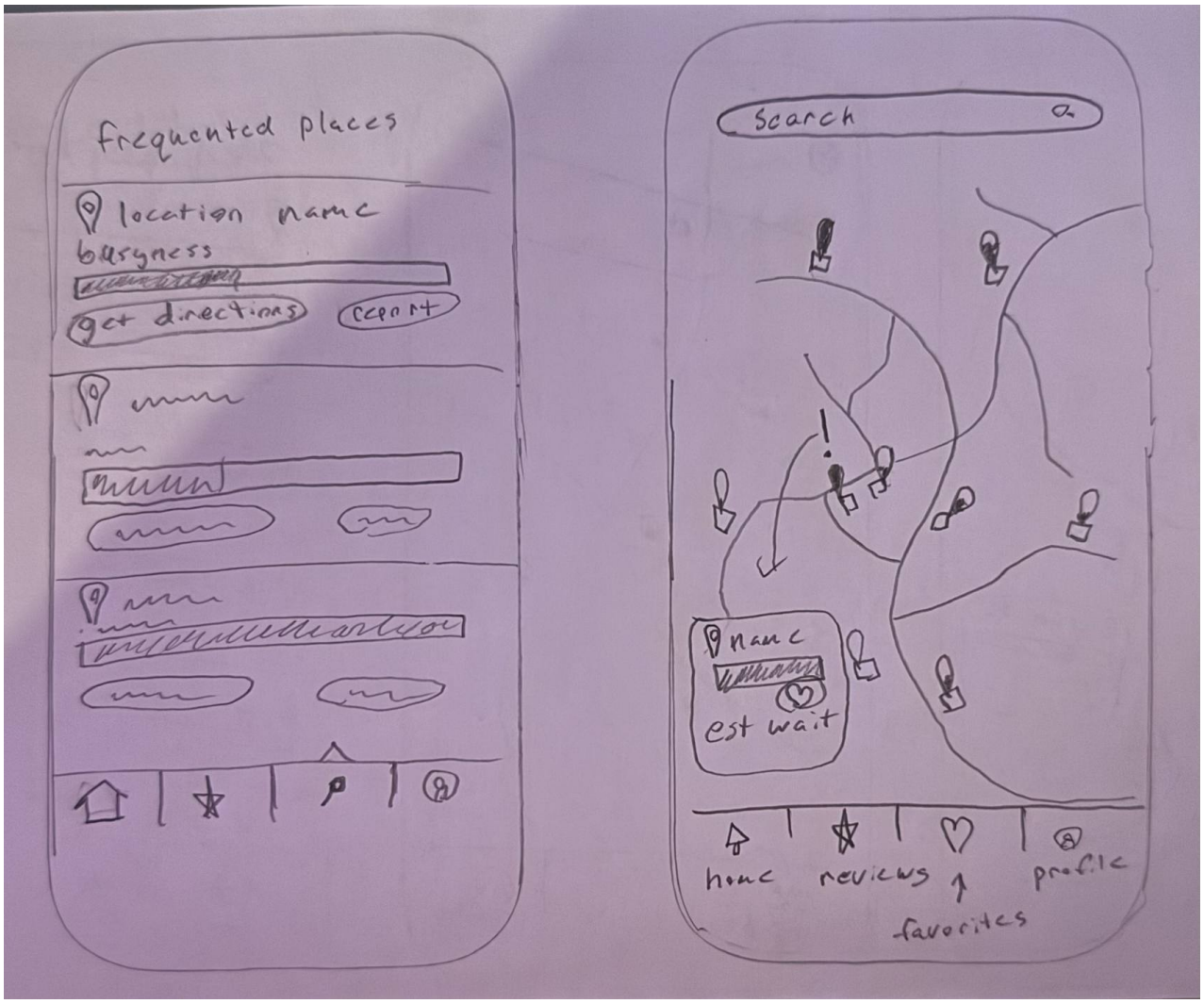
Group members: Nolan Jarvinen, Omar Naguib, Evan Wilkin

Step 1. Divergent Design Alternatives (Crazy Eights)

Nolan Jarvinen - Designs 1-8

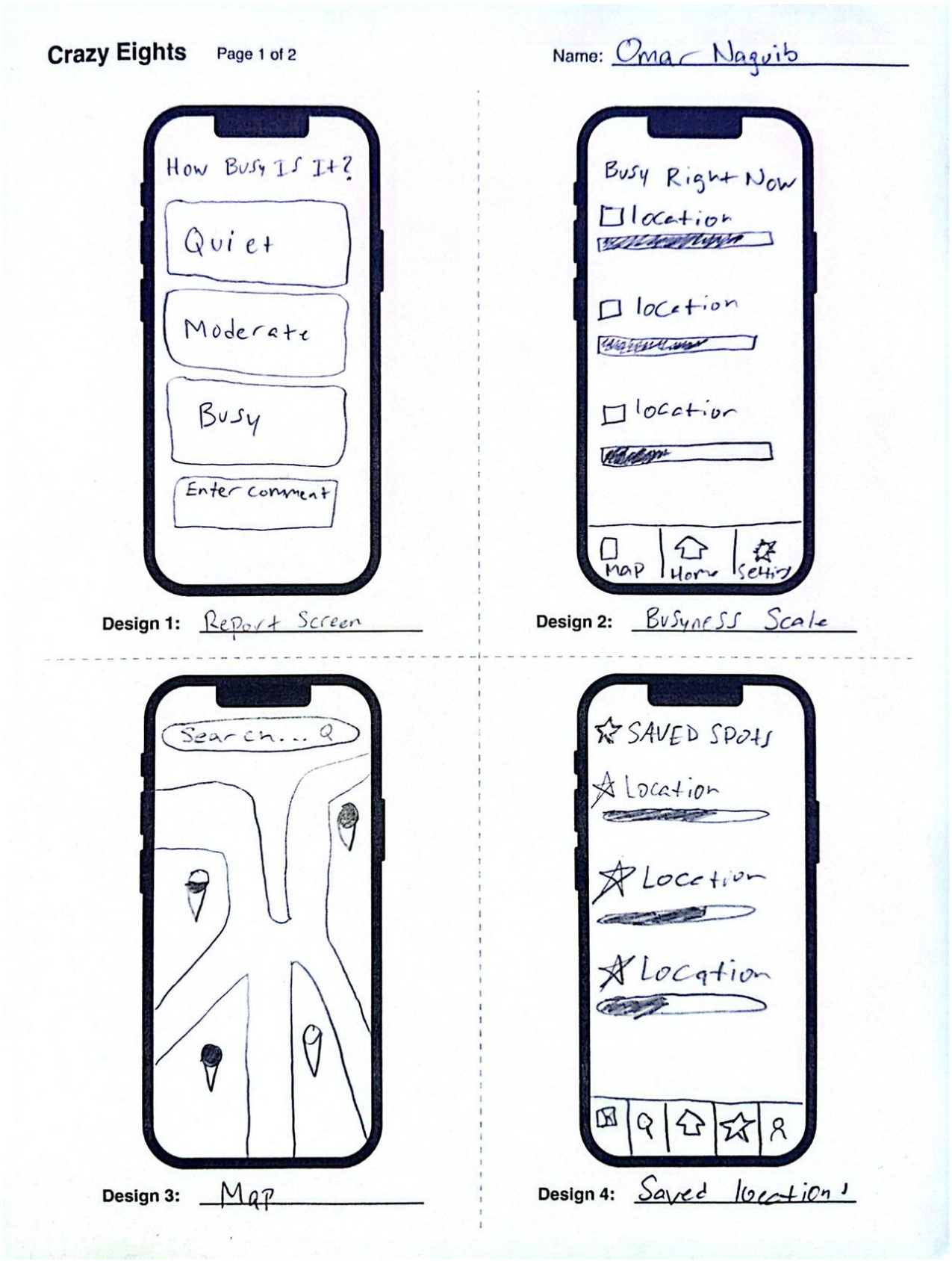


Nolan: sketches 1-6 (first sheet)



Nolan: sketches 7-8 (second sheet)

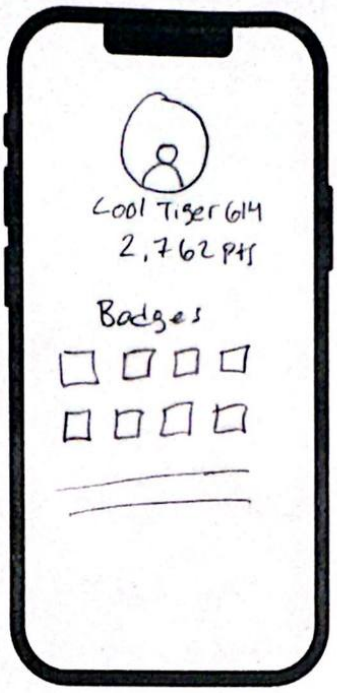
Omar Naguib - Designs 1-8



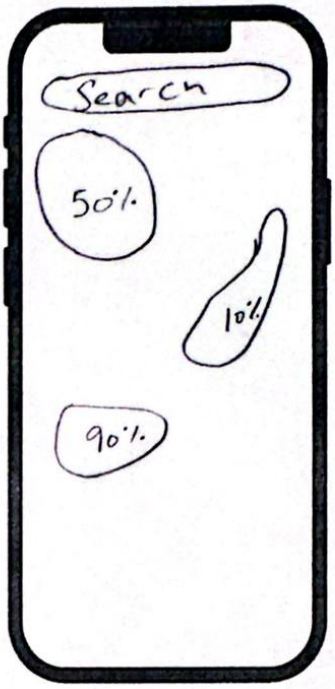
Omar: Crazy Eights, designs 1-4

Crazy Eights Page 2 of 2

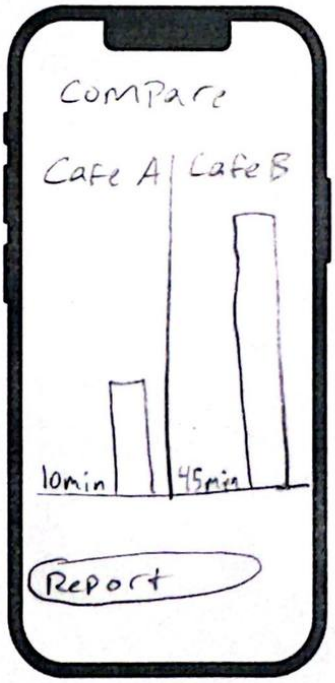
Name: Omar Nagvib



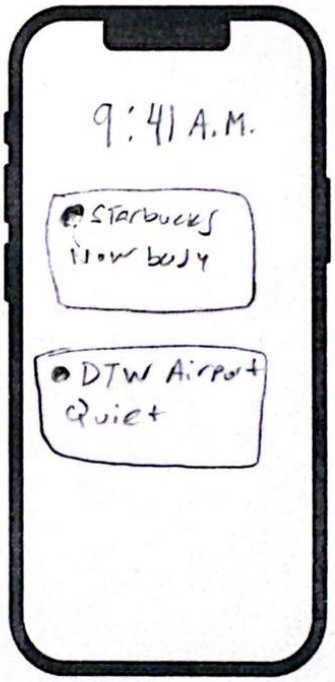
Design 5: Profile



Design 6: Heat Map



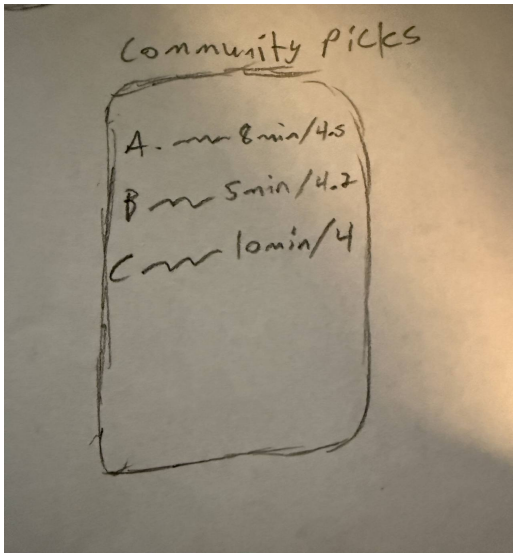
Design 7: Compare



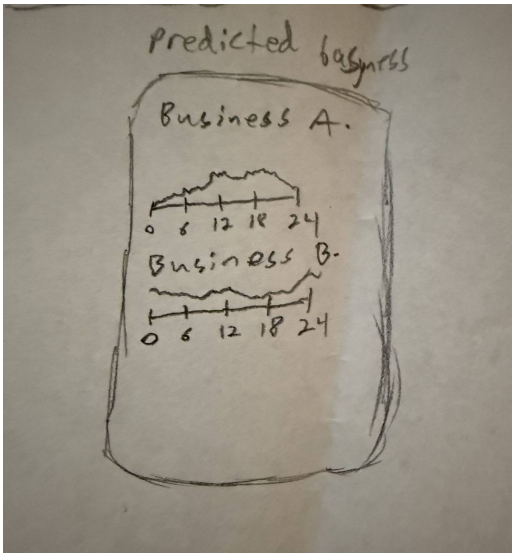
Design 8: Notification

Omar: Crazy Eights, designs 5-8

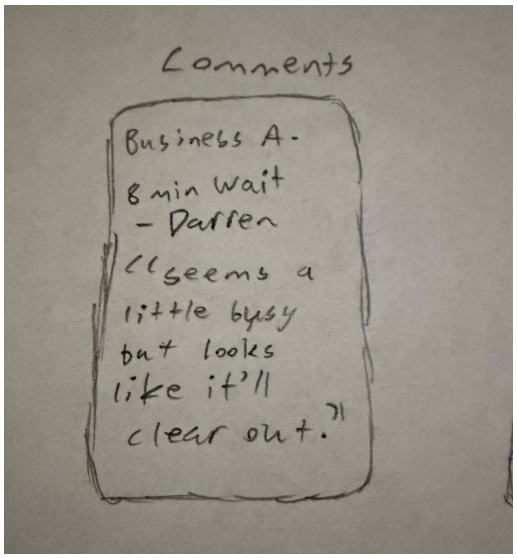
Evan Wilkin - Designs 1-8



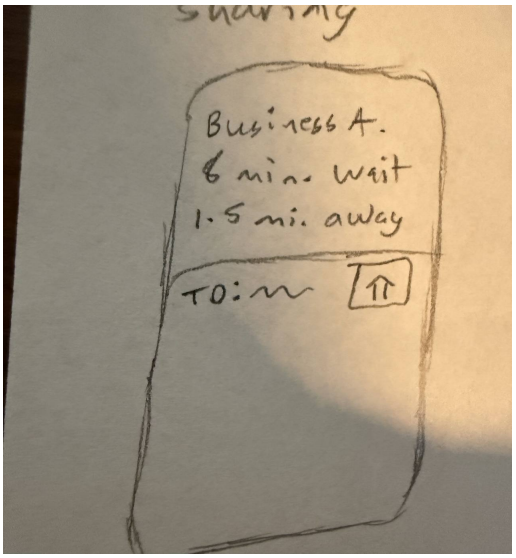
Design 1



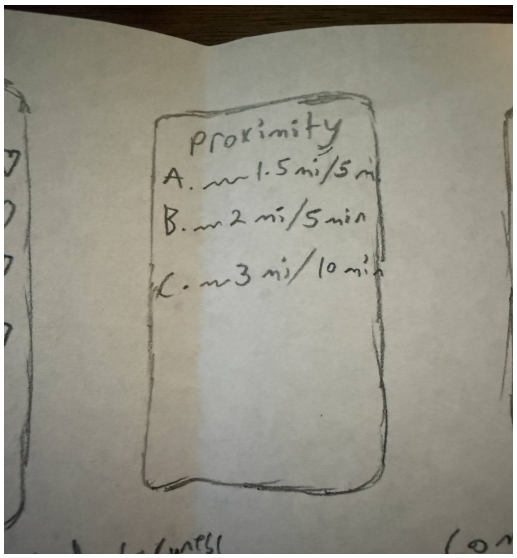
Design 2



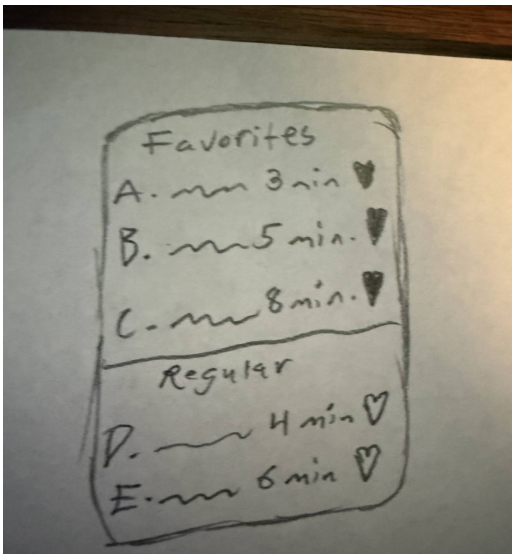
Design 3



Design 4



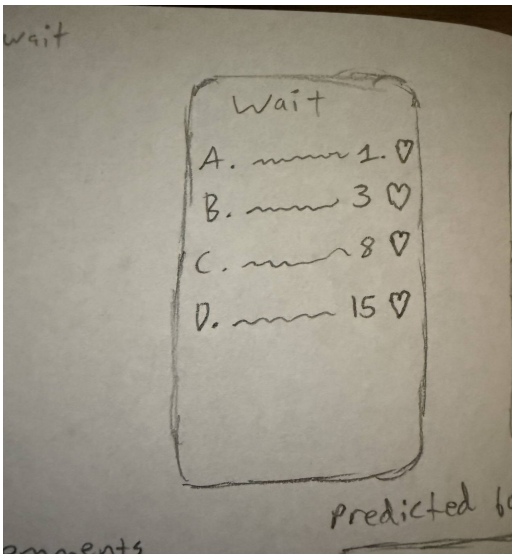
Design 5



Design 6



Design 7



Design 8

Step 2. Convergence and Final Design Sketches

After looking through everyone's sketches, our group decided to combine several ideas into one final design. We chose to make the map the main screen because we felt it would be the easiest way for users to find nearby businesses. We took inspiration from Omar's map, saved locations, and reporting designs, along with the nearby business listings and profile features from Nolan's sketches. We also incorporated ideas from Evan's sketches, including predicted busyness, community comments, wait times, and favorites.

We decided to include a business details page where users could see ratings, reviews, estimated wait times, and predicted busyness throughout the day. We also combined the different profile ideas to include achievements, activity streaks, and a user trust score. Although it wasn't included in our original sketches, we added a sign-in screen with a guest option because we felt accounts would be useful for saving favorites, leaving reviews, and tracking user activity.

The final design brings these features together while keeping navigation simple through the map and bottom navigation bar.

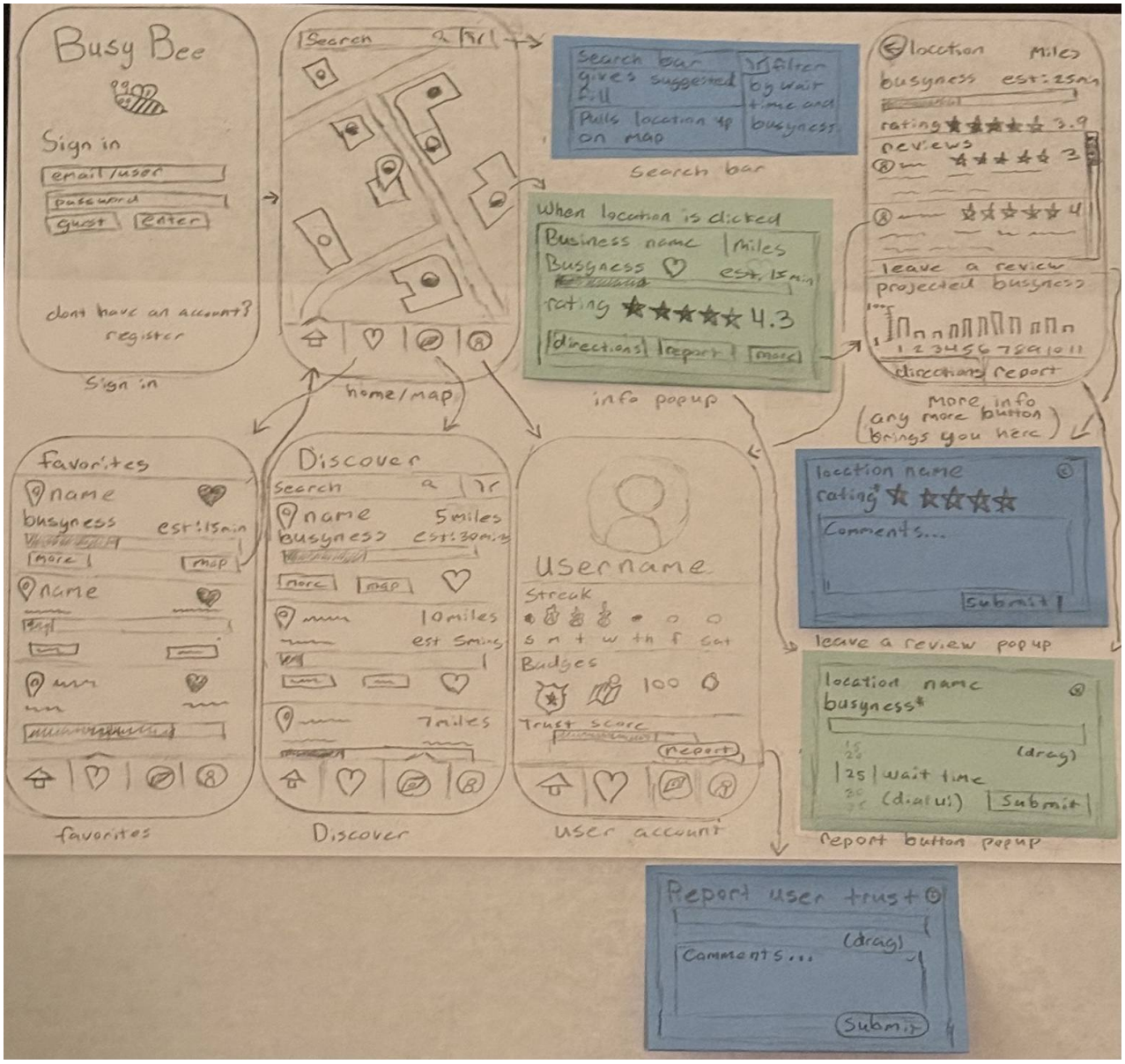


Figure 1. Final design flow map, with arrows showing navigation paths.

Step 3. Complete Paper Prototype

For our paper prototype, we took our flowchart and made a digital cover-up that removed the arrows and other navigation hints from our original design. Each group member was able to use this version of the prototype to conduct their cognitive walkthroughs without having the arrows guide them. This allowed us to go through the different tasks and see whether the buttons and navigation were clear enough for users to figure out on their own.



Figure 2. Paper prototype: screens and pop-ups without navigation arrows.

Step 4. Internal Cognitive Walkthrough

(made by Omar) 1. Report a Business's Busyness - Nolan Jarvinen (Jordan Ellis)
Select a business on the map > Select the Report button on the business pop-up > Move the busyness slider > Choose an estimated wait time > Submit report

(made by Evan) 2. Add a Business to Favorites - Nolan Jarvinen (Jordan Ellis)
Select a business on the map >Tap the Favorite button > Open Favorites to check the saved place

(made by Evan) 3. Check Predicted Busyness - Omar Naguib (Maya Whitfield)
Select business on map > Tap More to open details > View predicted busyness graph > Read predicted busyness graph

(made by Nolan) 4. Leave a Business Review - Omar Naguib (Maya Whitfield)
Select business > Open business details with More button > Select Leave a Review > Enter rating > Enter comment > Submit review

(made by Nolan) 5. View a Favorited Business on the Map - Evan Wilkin (Sam Okafor)
Open Favorites > Select a saved business > Tap Map button > View business location on map

(made by Omar) 6. Check and Report a User's Trust Score - Evan Wilkin (Sam Okafor)
Open a business's reviews > Select a user's profile picture > View their trust score > Submit a report

Task 1. Report a Business's Busyness

Nolan Jarvinen (Jordan Ellis)

Jordan makes plans on the spot, so submitting a report should be quick and easy.

Atomic action	Intent	Visibility	Association	Feedback	Proposed fix
Select a business on the map	Yes	Yes	Yes	Yes, the business popup opens.	None
Select the Report button on the business popup	Yes	Yes	Yes	Yes, the reporting popup opens.	None
Move the busyness slider	Yes	Yes	Partial. The slider has no clear scale.	Partial. It moves,	Show Quiet, Moderate, or Busy as the slider moves.
Choose an estimated wait time	Yes	Yes	Partial. There is a dial in the bottom left	Partial. The selected time stays in place on dial	Highlight the selection.
Tap Submit	Yes	Yes	Yes	No. There is no confirmation in the sketch.	Show a short Report submitted message.

Task 2. Add a Business to Favorites

Nolan Jarvinen (Jordan Ellis)

Jordan may want to save places while making plans with friends, so the heart button should be easy to find.

Atomic action	Intent	Visibility	Association	Feedback	Proposed fix
Select a business on the map	Yes	Yes	Yes	Yes, the business popup opens.	None
Tap the heart on the business popup to save the business	Yes	Yes	Yes. A heart usually means favorite.	Yes, the heart changes to a filled-in heart	None.
Open Favorites to check the saved place	Yes	Partial. The Favorites section itself is easy to find, but there is no way to search to find the one I just favorited instead, I have to scroll through all my favorites to find it	Yes	Partial.The Favorites list opens, but finding a recently saved business can be difficult when there are a lot of favorites.	Have the favorited business be searchable and filterable

Task 3. Check Predicted Busyness

Omar Naguib (Maya Whitfield)

Maya plans ahead and wants to know which day and time the forecast is showing.

Atomic action	Intent	Visibility	Association	Feedback	Proposed fix
Select a business on the map	Yes	Yes	Yes	Yes, the business popup opens.	None
Tap More to open details	Yes	Yes	Partial. More does not mention the forecast, so I wouldn't know it's in there.	Yes, the details screen opens.	Change More to Details or something that hints me in the right direction
Find the predicted busyness graph	Yes	Partial. It is below the reviews.	Yes. The section is labeled projected busyness.	Yes, the graph appears.	Put the graph above the reviews.
Read the prediction for a later time	Yes	Partial. The time labels are hard to understand.	No. The numbers 1 to 11 do not say what hours or day they mean.	Partial. She can see bars but cannot plan around them confidently.	Label times clearly, include the day, and briefly explain the prediction.

Task 4. Leave a Business Review

Omar Naguib (Maya Whitfield)

Maya pays attention to wait times and needs a clear way to tell a review apart from a busyness report.

Atomic action	Intent	Visibility	Association	Feedback	Proposed fix
Select a business on the map	Yes	Yes	Yes	Yes, the business popup opens.	None
Tap More to open details	Yes	Yes	Partial. More does not make the review option obvious.	Yes, the details screen opens.	Add a Review option to the business popup, or hint me to it with the more button
Tap Leave a Review	Yes	Partial. The link is down the page.	Yes	Yes, the review popup opens.	Make Leave a Review easier to see
Select a star rating	Yes	Yes	Yes	Yes, you can see all of the stars to click.	None
Enter a comment	Yes	Yes	Yes	Yes, the comment can be seen in the field.	None
Tap Submit	Yes	Yes	Yes	No. There is no posting confirmation.	Show the posted review and display the new review.

Task 5. View a Favorited Business on the Map

Evan Wilkin (Sam Okafor)

Sam likes to plan ahead and avoid crowded places, so being able to quickly check his saved businesses is important to him.

Atomic action	Intent	Visibility	Association	Feedback	Proposed fix
Open Favorites	Yes	Yes. The bottom navigation makes Favorites easy to find.	Yes. Sam expects to see his saved businesses.	Yes. His favorites list appears.	None.
Select a saved business	Yes	Yes. The saved businesses are listed clearly.	Yes. Sam sees the listed businesses and understands they are his favorited businesses	Partial. Businesses are shown, but it's hard to find what Sam is looking for.	Make Favorites searchable so Sam wouldn't have to scroll to find the business he’s looking for.
Tap the Map button	Yes	Yes. The Map button is under the business name.	Yes, the Map button brings you to the live map	Yes. The map opens.	Let users open the map by tapping the saved business directly.
View business location on map	Yes	Yes. The location and business information are visible.	Yes. Sam can check the location and busyness.	Yes. The business popup shows, and the map is centered on the location	Keep the More option available in the map popup.

Task 6. Check and Report a User's Trust Score

Evan Wilkin (Sam Okafor)

Sam generally trusts information shared by other users. If he reported someone, he would want to know the report was received and whether anything was done about it.

Atomic action	Intent	Visibility	Association	Feedback	Proposed fix
Open a business's reviews	Yes. Sam wants to check reviews and recent wait reports.	Partial. Reviews are under More instead of having their own section.	Yes. Reviews lead to user feedback and predicted busyness information.	Yes. Reviews appear.	Make reviews easier to reach from the business popup.
Select a user's profile picture	Yes. Sam wants to see who posted the information.	Yes. The profile picture is easy to select.	Yes. Selecting the picture opens that user's profile.	Yes. The profile opens.	None.
View their trust score	No. Sam generally assumes people are being honest about busyness, so he may not think to check a trust score.	Yes. The trust score is easy to see on the profile.	Yes. The score is clearly connected to the user.	Yes. Sam can see the trust score.	Briefly explain what the trust score means and how it is calculated.
Submit a report	Yes. If Sam received bad information, he would want to warn others.	Yes. Report is next to the trust score.	Yes. Report lets Sam rate the user's trustworthiness and explain why	Partial. Sam doesn't know if it went through or what happens after reporting.	Show a report confirmation and explain to the user why their report matters.